PRIME FACTOR STUDIO — IN-BOX USER GUIDE

Kinesthetic Number Theory, Proportional Logarithmic Stacking & 12-Socket Lid Tray

SERIES 1 • CARD 05 / 12 • DOMAIN: FUNDAMENTAL THEOREM OF ARITHMETIC

CARD 05 / 12 • PRIME-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

FEEL THE PRIMES. MULTIPLY WITH HEIGHT.

Standard base-10 counting blocks add lengths (1+1=2).

Prime Factor blocks are logarithmically proportional:

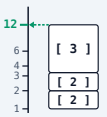
log(a × b) = log(a) + log(b)

Stacking blocks in a tower multiplies their numbers!

CALIBRATED LOGARITHMIC DATUM

- Unit Datum: Block [2] = 1U = 7.00 mm visible height
- Log Modulus: K_log = 23.2535 mm / decade
- Prime Formula: h_vis(p) = K_log × log10(p)

VISUAL PROOF: TOWER 12 = [2] + [2] + [3]



▲ SUMMIT = 12

2 × 2 × 3 = 12

Visible: 25.09 mm

Commutative:

[3]+[2]+[2] = same height!

2. KIT PARTS INVENTORY (23 PIECES)

COMPONENT	QTY	SPEC	COLOR
Prime Block [2]	2x	7.00 mm	Matte Blue
Prime Block [2²]	2x	14.00 mm	Matte Blue
Prime Block [2³]	2x	21.00 mm	Matte Blue
Prime Block [3]	4x	11.09 mm	Muted Red
Prime Block [3²]	1x	22.18 mm	Muted Red
Prime Block [5]	2x	16.25 mm	PLA White
Prime Block [7]	2x	19.65 mm	PLA Gray
Prime Block [11]	1x	24.21 mm	PLA Tan
Prime Block [13]	1x	25.90 mm	PLA Tan
Prime Block [17]	1x	28.61 mm	PLA Gray
Prime Block [19]	1x	29.73 mm	PLA Gray
Prime Block [23]	1x	31.67 mm	PLA White
Prime Block [29]	1x	33.51 mm	PLA White

TOTAL: 21 Blocks + 7U Box + Stage Lid = 23 Parts (7U Farm) • 47 Parts (13U)

SLANT 3D PRODUCTION PALETTE

- PLA Matte Blue: Prime [2] family (2, 2², 2³).
- PLA Matte Muted Red: Prime [3] family (3, 3²).
- PLA White: Prime [5], Primes [23], [29], Enclosure Box & Lid.
- PLA Tan: Mid primes [11], [13] • PLA Gray: Primes [7], [17], [19].

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Stand prime blocks in product-labeled sockets (2^{12}=4096, 3^7·2=4374, etc). All 12 towers level off at the 84mm waterline. Snap stage lid flush for silent carry.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

Every integer > 1 has a unique prime factorization decomposition.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: HOW TO FIND THE LEAST COMMON MULTIPLE (LCM)

Step-by-Step Hands-On Tutorial: Finding LCM(12, 18) = 36 & The Law of Conservation of Primes

SERIES 1 • CARD 05 OF 12 • DOMAIN: FUNDAMENTAL THEOREM OF ARITHMETIC

CARD 05 / 12 • PRIME-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/prime-studio/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 6.NS.B.4 (LCM & GCD)
- CCSS 4.OA.B.4 (Factor Pairs)
- TEKS 111.26.B.6.7.A (Prime Factor...)

Grades 4–9 Math & Pre-Algebra

GUIDING INQUIRY

Which prime blocks are shared by both towers, and which blocks are unique to one tower?

1 STEP 1: FACTOR & BUILD TOWERS 12 & 18

Decompose numbers 12 and 18 into prime building blocks:

- Tower 12: Stack [2] + [2] + [3] (2 × 2 × 3 = 12)
- Tower 18: Stack [2] + [3] + [3] (2 × 3 × 3 = 18)

2 STEP 2: FIND SHARED OVERLAP (THE GCD)

Stand Tower 12 and Tower 18 side-by-side. Identify the blocks they share in common (intersection):

- Tower 12 has: [2], [2], [3] I Tower 18 has: [2], [3], [3]

Shared Overlap: Exactly one [2] and one

3-4 STEPS 3 & 4: ASSEMBLE THE LCM TOWER

3. TAKE PRIME UNION: Take highest power of each prime: two [2]s, two [3]s.

4. STACK LCM TOWER: [2] + [2] + [3] + [3] = 36 (Summit: 36.19 mm)!

Calculator Verification: Press Factor Tree Icon ([PRIME] / [g][FACT]) and run 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [f] [LCM] → 36.

5 CORE PRINCIPLE: CONSERVATION OF PRIMES: A × B = GCD × LCM

Multiply Tower 12 and Tower 18: you have three [2]s and three [3]s. Look at your GCD Tower ([2], [3]) and your LCM Tower ([2]², [3]²): together they contain the EXACT same prime blocks!

Therefore: A × B = GCD(A, B) × LCM(A, B) = 6 × 36 = 216!

★ EXTENSION INVESTIGATION: FIND LCM(8, 12)

Build Tower 8: [2]+[2]+[2] and Tower 12: [2]+[2]+[3]. Prime union takes three [2]s and one [3] → LCM = 24 (28.09mm)! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem • Mountain View, CA • BPA-Free

PARTS & REFILL PORTAL

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STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH

Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC

Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY

Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING

Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5

Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC

[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS

Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS

Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING

Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1

Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION

Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS

Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED

Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS

Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4

Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:

Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:

Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR STORAGE DETENTS:

Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:

Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

PRIME-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:

All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:

Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] ZERO-RATTLE DETENT LOCK:

Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:

2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:

100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:

Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 23 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io

stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

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